WORKED SOLUTION

STEP II Specimen paper, Question 6

Pure • Functional equation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s2-q6

The key idea

We are not allowed to say "f is e^x", so everything has to come from the differential equations themselves. The one tool we may use is that a function whose derivative is zero everywhere is constant. So the trick is to build an expression whose derivative cancels to zero, and then pin down the constant by putting $x = 0$. Pairing the value at $a - x$ with the value at x is natural because the two arguments always add up to a, which is exactly the shape of an addition formula.

Solution

The exponential addition law

Fix a and let $p(x) = \mathrm{f}(a - x)\,\mathrm{f}(x)$. By the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x}[\mathrm{f}(a - x)] = -\mathrm{f}'(a - x) = -\mathrm{f}(a - x)$, using $\mathrm{f}' = \mathrm{f}$. So by the product rule

$$\begin{aligned} p'(x) &= -\mathrm{f}(a - x)\,\mathrm{f}(x) + \mathrm{f}(a - x)\,\mathrm{f}'(x) \\ &= -\mathrm{f}(a - x)\,\mathrm{f}(x) + \mathrm{f}(a - x)\,\mathrm{f}(x) = 0, \end{aligned}$$

which is the first result.

Since $p'(x) = 0$ for every x, the function p is constant. Its value at $x = 0$ is $\mathrm{f}(a)\,\mathrm{f}(0) = \mathrm{f}(a)$, since $\mathrm{f}(0) = 1$. Hence

$$\mathrm{f}(a - x)\,\mathrm{f}(x) = \mathrm{f}(a) \qquad \text{for all } x \text{ and all } a.$$

Now take any x and y and choose $a = x + y$. Then $a - x = y$, and the identity becomes

$$\mathrm{f}(y)\,\mathrm{f}(x) = \mathrm{f}(x + y),$$

as required.

The cosine addition formula

We copy the same strategy, so we need a combination of c and s, evaluated at $a - x$ and at x, whose derivative vanishes.

Ask yourself Which combination should we differentiate?

Let the answer guide us: with $a = x + y$ we want $\mathrm{c}(x)\mathrm{c}(y) - \mathrm{s}(x)\mathrm{s}(y)$, that is, $\mathrm{c}(a - x)\mathrm{c}(x) - \mathrm{s}(a - x)\mathrm{s}(x)$. So that is the function to try.

Fix a and let

$$g(x) = \mathrm{c}(a - x)\,\mathrm{c}(x) - \mathrm{s}(a - x)\,\mathrm{s}(x).$$

By the chain rule and the given derivatives,

$$\frac{\mathrm{d}}{\mathrm{d}x}[\mathrm{c}(a - x)] = -\mathrm{c}'(a - x) = \mathrm{s}(a - x), \qquad \frac{\mathrm{d}}{\mathrm{d}x}[\mathrm{s}(a - x)] = -\mathrm{s}'(a - x) = -\mathrm{c}(a - x).$$

Hence, by the product rule,

$$\begin{aligned} g'(x) &= \big[\mathrm{s}(a - x)\,\mathrm{c}(x) - \mathrm{c}(a - x)\,\mathrm{s}(x)\big] - \big[-\mathrm{c}(a - x)\,\mathrm{s}(x) + \mathrm{s}(a - x)\,\mathrm{c}(x)\big] \\ &= 0. \end{aligned}$$

So g is constant, and its value is

$$g(0) = \mathrm{c}(a)\,\mathrm{c}(0) - \mathrm{s}(a)\,\mathrm{s}(0) = \mathrm{c}(a) \cdot 1 - \mathrm{s}(a) \cdot 0 = \mathrm{c}(a).$$

Therefore $\mathrm{c}(a - x)\mathrm{c}(x) - \mathrm{s}(a - x)\mathrm{s}(x) = \mathrm{c}(a)$ for all x and a. Given any x and y, put $a = x + y$, so that $a - x = y$:

$$\mathrm{c}(x + y) = \mathrm{c}(x)\,\mathrm{c}(y) - \mathrm{s}(x)\,\mathrm{s}(y),$$

which is the required result.

The same method with $h(x) = \mathrm{s}(a - x)\,\mathrm{c}(x) + \mathrm{c}(a - x)\,\mathrm{s}(x)$ gives $h' = 0$ and $h(0) = \mathrm{s}(a)$, so $\mathrm{s}(x + y) = \mathrm{s}(x)\mathrm{c}(y) + \mathrm{c}(x)\mathrm{s}(y)$ too.

Checks

- Put $y = 0$: the results reduce to $\mathrm{f}(x) = \mathrm{f}(x)$ and $\mathrm{c}(x) = \mathrm{c}(x)$, using $\mathrm{f}(0) = \mathrm{c}(0) = 1$ and $\mathrm{s}(0) = 0$.
- Put $y = -x$ in the cosine result: $1 = \mathrm{c}(0) = \mathrm{c}(x)\mathrm{c}(-x) - \mathrm{s}(x)\mathrm{s}(-x)$, which is $\cos^2 x + \sin^2 x = 1$ in disguise.
- We solved the differential equations numerically (a Runge–Kutta method, with no built-in exponential or trigonometric functions) and tested both identities at 60 random pairs in $[-3, 3]$; for example, with $x = 0.7$ and $y = 1.1$ both sides of the cosine result are -0.2272 to four decimal places. All differences were below 10^{-11}.

Common slips

- Forgetting the minus sign from the chain rule when differentiating $\mathrm{f}(a-x)$, $\mathrm{c}(a-x)$ or $\mathrm{s}(a-x)$.
- Stopping at "the derivative is zero" without evaluating the constant at $x=0$ and then choosing $a=x+y$.
- Quoting $\cos(x+y)$ or $\mathrm{e}^{x+y}=\mathrm{e}^{x}\mathrm{e}^{y}$, which the question explicitly forbids.

Beyond STEP

At university this is the honest way to define the exponential and trigonometric functions: as solutions of differential equations (or as power series), with every familiar property deduced from that definition. The fact we used, that zero derivative everywhere forces a function to be constant, is a consequence of the Mean Value Theorem, and it also shows these solutions are unique. Abbott's *Understanding Analysis* builds the theory carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.